

Richard J Gresham

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SUMMARY

Computer Science graduate (GPA 3.96) with strong experience in C++ software development, object-oriented design, multithreading, and data structures. Proven ability to design, implement, and debug full-lifecycle software systems through academic and personal projects. Seeking an early-career Software Engineer role in systems, embedded, or application development. U.S. citizen, eligible for U.S. Secret security clearance.

SKILLS

- **Languages:** C++, C, Python, SQL
- **Core Concepts:** Object-Oriented programming, Data Structures, Algorithms, Multithreading, Concurrency, UML, Agile/Scrum
- **Tools:** Git (version control), Linux, Standard Library/ STL, Visual Studio, Jira, Android Studio
- **Frameworks:** Qt, MFC, Pygame

SOFTWARE ENGINEERING PROJECTS & EXPERIENCE

Huffman Encoding/Decoding app | C++, Qt

Oct 2025 - Dec 2025

- Developed a C++/Qt desktop application to compress and decompress text files using Huffman coding with a graphical file-selection interface.
- Implemented a custom priority-queue-based Huffman tree algorithm for generating optimal prefix codes, achieving 40-60% file reduction on large text inputs.
- Designed file validation and error handling to ensure reliable execution.

Peer to Peer Chat application | C++, MFC

Jun 2025 - Jul 2025

- Built a multithreaded peer-to-peer chat application enabling real-time LAN messaging using C++ socket programming.
- Implemented concurrent send/receive threads to prevent UI blocking and ensure responsive communication.
- Developed GUI components for message display and user input using Microsoft Foundational Classes (MFC).
- Added configurable IP and port selection to support flexible peer connection.

Alpha-Beta Pruning Chess Engine | Python, Pygame

Jan 2023 - May 2023

- Led a 6-member Agile team as Scrum Master to design and implement a chess engine using alpha-beta pruning.
- Developed custom-heuristic evaluation function based on research, improving AI move quality.
- Optimized search depth constraints, increasing AI decision making speed by over 40%.
- Implemented interactive UI elements, animations, and audio feedback for real-time gameplay.

EDUCATION

California State University, Fullerton

Aug 2020 - May 2023

Bachelor of Science, Computer Science

- **GPA:** 3.96 / 4.0

Golden West College

Sep 2017 - May 2020

Associate of Science, Computer Science

- **GPA:** 3.94 / 4.0

HONORS AND AWARDS

- **Summa Cum Laude:** California State University, Fullerton (2023)
- **ACM OC Awards for Excellence in Computing Sciences:**(2023)